

The Palm™ Debugger: Taking Control

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More connected.

Overview

- How to get connected
- There is a debugger mode and console mode
- How to inspect the device
  - Disassembly
  - Stack trace
- How to manipulate databases
  - Import
  - Export
  - Delete

PalmDebugger

PalmSource 99

- For when there's no source code
  - Palm OS® software and other applications
  - No source level debugging
- For when you want to inspect memory heaps
- For importing and exporting databases
- Macintosh™ or Windows NT™ platforms

Connecting

- Connect to device
  - Serial connection
  - 57K is the default baud rate
  - Type shortcut… 1 on device
- Connect to Palm OS® Emulator
  - Internal connection
  - No baud rate necessary

Debugger Mode

- The window on the left
- Halts the CPU
- Allows inspection of machine
  - Disassembly
  - Registers
  - Stack trace
  - Memory display

Attach

- Start talking to the device

```
att
Remote stopped due to: DEBUGGER TRAP EXCEPTION
'Search'
  +$0064  00025FE0  *TST.B     -$0035(A5)
```

Stack Crawl

- To see what function the program is in

```
sc
Calling chain using A6 Links:
 A6 Frame   Caller
 0001BA2E  10C0FBE2  SysHandleEvent+0010
 0001B87A  10C65E3E  Find+0118
 0001B7B0  10C65968  FindSearch+027A
 0001B740  10C11196  SysAppLaunch+04D0
 0001B6BC  00025A7C  __Startup__+0D42
 0001B69C  0002D42E  PilotMain+0020
```

Instruction List

- To see what instructions the program was executing

```
il pc-20
'Search  00025F7C'
 +$0052  00025FCE   _DmOpenDatabase
 +$0056  00025FD2   MOVE.L     A0,D5
 +$0058  00025FD4   ADDQ.W     #$08,A7
 +$005A  00025FD6   BNE.S      Search+$0064
 +$005C  00025FD8   CLR.B      $0004(A4)
 +$0060  00025FDC   BRA.W      Search+$0216
 +$0064  00025FE0  *TST.B      -$0035(A5)
 +$0068  00025FE4   BEQ.S      Search+$0070
```

Registers

- To see the values of the registers

```
reg
D0 = 0001CC68     A0 = 0001CE04     USP = 00000000
D1 = 00002000     A1 = 10C8C106     SSP = 0001B5BE
D2 = 00000001     A2 = 000024AE
D3 = 00000000     A3 = 00000000
D4 = 00000001     A4 = 000024AE
D5 = 0001CE04     A5 = 8001B73C
D6 = 00010000     A6 = 0001B668     PC  = 00025FE0
D7 = 00020000     A7 = 0001B5BE     SR  = tSxnzvc
```

Heap List

- To get a list of valid heaps and memory addresses

```
hl 0
idx  hpID  hpPtr   size    free      maxFree    flags
-----------------------------------------------------
0    00 000015B0 0001EA50 0016FB4  016E30      4000
1    01 0002010E 001DFEF2 01DABBE  1DA36E      4000
2    02 10C08212 00137DEE 000573C  005734      4001
```

Heap Dump (Dynamic Heap)

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- To explore memory usage

```
hd 0
Heap Summary:
  flags:                  4000
  size:                   01EA50
  numHandles:             #200
  Free Chunks:            #11          (016DB0 bytes)
  Movable Chunks:         #6           (000CBA bytes)
  Non-Movable Chunks:     #44(006CB2 bytes)
  Owner  0:                      01E0C0 bytes
  Owner  2:                      00065C bytes
```

Heap Dump (Dynamic Heap)

- To track memory leaks

```
hd 0
 start          size  lock owner flags type
-----------------------------------------------
-00001B38   00022 #0  #0   fM Alarm Table
-00001BFC   00456 #0  #0   fM Graffiti Private
 000020E2   18FFA #0  #0   FM largest free chunk
•0001B0E4   009C0 #15 #0   fm App Globals
•0001BAAC   00200 #15 #0   fm
*0001BCB4   00044 #1  #2   fM Form 1900
 0001BD00   00030 #0  #0   FM --> 0001BDA4
```

Heap Dump (Storage Heap)

- To find records or app info block

```
hd 1
start      size lock owner flags type index attr
-----------------------------------------------
*000208E6 000050 #1   #1    fM   DBHdr AddressDB
-00020964 00027E #0   #2    fM   DBAID AddressDB
-00021C6E 000016 #0   #2    fM   Rcrd  #0 xDbs #2 AddressDB
-00021D84 000022 #0   #2    fM   Rcrd  #1 xDbs #2 AddressDB
```

Opened Databases

- Databases listed in reverse order

opened

```
name       resDB  crdNum   acessP    opnCnt  mode
-----------------------------------------------
 AddressDB     no     0  0001AD0C     1        0013
*Address Book  yes    0  0001FB92     1        0001
*Graffiti      yes    0  0001FC68     1        0007
*System        yes    0  0001FFEE     1        0005
-----------------------------------------------
Total: 4 databases opened
```

Database Directory

- Lists all databases

```
dir 0
name              ID         total          data
-----------------------------------------------
*Address Book  10D3F08A    57.6 Kb        56.9 Kb
*Calculator    10D3EE8E    14.5 Kb        13.7 Kb
*Date Book     10D3EB48   106.9 Kb       105.5 Kb
 AddressDB     00020147     0.1 Kb         0.0 Kb
 Sys MIDI Snds 00020187     1.0 Kb         0.8 Kb
```

Displaying Memory

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- Display a memory as bytes
- Remember it's big-endian

```
hd 0
*00001F46 000304 #1  #2  fM  Form "Address  List"

dm 00001F46

1F46: 00 A0 00 A0 00 01 B1 D4
      1A 00 00 00 00 00 00 A0
".†.†..±‘.......†"

dm 00001F46 000304
```

Displaying Memory

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- Display memory as a structure

```
hd 0
*00001F46 000304 #1  #2  fM  Form "Address  List"

dm 00001F46 FormType

00001F46  struct    FormType
            {
00001F6E    UInt16  formId    = $3E8
00001F70    UInt16  attr      = $A400
00001F72    WinPtr  bitsBehindForm  = $00000000
00001F76    MemPtr  handler   = $000010CB
```

Setting Values

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- Set a value in memory
  - `sb <addr> <value>`
- Set a register
  - `d0=0`
  - `pc=10C7C23A`

Math


```
00016EDA
$00016eda  #93914  #93914  '..n.'
000015B0+0001EA50
$00020000  #131072  #131072  '....'

a6
$0001ead6  #125654  #125654  '....'
dm a6
0001EAD6: 00 01 EA E0 10 C0 F4 0A
dm a6+4
0001EADA: 10 C0 F4 0A 00 00 00 01
dm @a6
0001EAE0: 00 01 A0 94 10 C7 AB 8C
```

Conversions

- Hex, decimal, and ASCII are supported

```
41
$41  #65  #65  'A'
#65
$00000041  #65  #65  '...A'
'A'
$41  #65  #65  'A'
```

Resuming

- `g`
  - Resumes the CPU
  - The opposite of att
- `gt <addr>`
  - Go until the address is reached
  - Useful for getting past long loops

Stepping

- `s` step into routines
- `t` step over routines

Watchpoints

- Watch four bytes and stop when they change
- `ss <addr>`
  - `ss 0` to see when NULL is written to
- Slow but effective

API Breakpoints

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- `atb “EvtGetEvent”`
  - Stop when a particular API is called
  - This stops every time an event is requested
- `atd` to display all breakpoints
- `atc` <addr> to clear a breakpoint

Breakpoints

- `br` for a break point
  - Get an address from sc or il
- `brd` to display
- `cl` or `brc` to clear

Finding

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- `fb <value> <addr> <number of bytes>`
- `ft <text>  <addr> <number of bytes>`
  - Use hl to find out where heaps are to search them
  - `ft “roger” 000015B0 0001EA50`
  - Use to find a routine’s address

More Stack Crawl

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- sc6 [<addr> [<frames>]]
  - `sc a6 50`
- Try Sc7 when sc doesn't work

Talking to the Debugger

- DbgBreak()
  - Treated like a breakpoint
- DbgMessage()
  - Sends text to the debugger via the serial port
  - Doesn’t stop

Console Mode

- The right window of the debugger
- Used for changing data on the device
- The device keeps running
- Enter by writing shortcut… 2
  - Leaves the serial port open
  - Drains more power than normal
  - HotSync® communications are not possible
  - Reset the device to clear console mode

Dir

- More interesting than the debugger version
- Number of records
- Attributes
- Version
- Creation Date
- Backup Date
- Modification Number
- App Info handle
- Type and Creator

Standard Heap Routines

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- Heap List
- Heap Dump

Heap Fill

- Great way to fill up the storage heap and test low memory conditions
- hf <hex heapID> [free bytes]
  - `hf 1 400`

Importing and Exporting Databases

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- Import 0 “Reptoids.prc”
- Export 0 “Saved Preferences”
- ExportAll 0 “Roger”
- ImportAll 0 “Roger”

Delete Databases

- Del 0 “Reptoids.prc”
- Can’t delete opened databases
  - Leave the app
  - Close the database
    - Opened

```
opened
name       resDB  crdNum   acessP   opnCnt  mode
*Address Book  yes   0   0001FB92     1     0001
```

    - Close

```
close 0001FB92
```

Deleting Resources

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- DelResource <access ptr> <index>
- Open

```
open 0 “Saved Preferences”
Success! Access Pointer = 0001A578
```

- ListResources

```
listresources 0001A578
index   localID     size       type     id  data
  0     0002012B     52        'psys'    0
  1     00020257      4        'irda'    0
```

- Close

Memory Debugging

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- Useful for the device, not Palm OS™ Emulator
- Mdebug
- Fill chunks when freed
- Move unlocked chunks
- Check the heap

Capturing the Output

- History
- Searching, especially through heap dumps
- Select All, Copy, Paste
- Log filename

Done with the Console

- Reset
- Closes the serial port

Help

- Both the debugger and console have help
- Help lists all the commands
- Help can be requested for each function

```
help del
Delete a Database.
Syntax: del <cardNo> <databaseName>
```

Summary

- How to see in which function a problem is
- How to see if registers read from valid locations
- How to inspect the heaps
- How to examine and walk through code
- How to find, import and export databases


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